Neural Signaling

MED 808: Neuroscience, Neuroanatomy, and Neurology (NNN); Fall 2026

Reading:

Linda S. Costanzo: Physiology, 8th Ed., "Refer to Course Syllabus".

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Learning Objectives

After studying this lecture, the student should be able to:

1. Classify nerve fibers by conduction velocity, diameter, and myelination, and explain how these properties affect signal transmission.
2. Identify the types of sensory receptors and explain how excitation of a sensory nerve fiber by a receptor potential produced in a Pacinian corpuscle or Rod cell generates a receptor potential.
3. Describe how different types of sensory receptors transduce external stimuli into neural signals, including key mechanisms of receptor activation.
4. Explain the concept of sensory adaptation and its physiological significance in normal perception and clinical conditions.

Lecture Outline

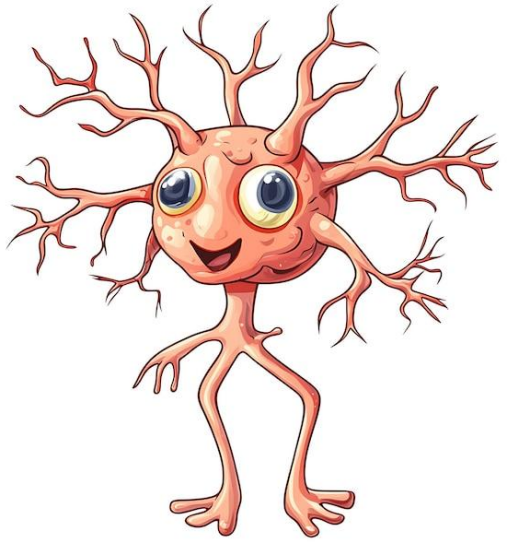
I. Types of Nerve Fibers

II. Sensory Reception

III. Sensory Transduction and Receptor Potentials

IV. Types of Sensory Receptors and Their Transduction Mechanisms

V. Sensory Adaptation



Types of Nerve Fibers

Overview of Nerve Fibers

- **Nerve fibers** are the axons of neurons that carry electrical impulses.
- **Sensory and Motor Nerve Fibers**
  - **Sensory Fibers (Afferent):** Transmit sensory information from receptors (e.g., touch, pain, proprioception) to the CNS
  - **Motor Fibers (Efferent):** Carry motor commands from the CNS to muscles and glands for actions (voluntary and involuntary)
- **Nerve fibers** differ in diameter, conduction velocity, and function.

Nerve Fiber Classification, Cont.

- **Based on Diameter and Conduction Velocity:**
  - **A fibers** (myelinated, fast- to medium- conducting)
  - **B fibers** (moderately myelinated, slower-conducting)
  - **C fibers** (unmyelinated, slow-conducting)
- **Based on Function:**
  - Sensory fibers (afferent)
  - Motor fibers (efferent)
  - Autonomic fibers (sympathetic and parasympathetic)

A Fibers

- **A Fibers:** Myelinated
- **Types of A fibers:**
  - **Aα (alpha):** Motor fibers for muscle control and proprioception; **(large-diameter); fast conducting**
  - **Aβ (beta):** Sensory fibers for touch, pressure, and vibration; **(large-diameter); fast conducting**
  - **Aγ (gamma):** Motor fibers for muscle spindles (regulating muscle tone); **(medium-diameter); medium conducting**
  - **Aδ (delta):** Sensory fibers for touch, pressure, **fast pain,** and temperature (e.g., cold); **(small-diameter); medium conducting**
- **Conduction Velocity:** 80-120 m/sec (Aα) to 5-30 m/sec (Aδ).

B Fibers

- **B Fibers:** Myelinated, Slower Conduction
  - Found mainly in the autonomic nervous system **(e.g., preganglionic autonomic nerves)**
  - Carry signals from the CNS to autonomic ganglia (preganglionic fibers)
  - Conduction velocity is slower than A fibers (3-15 m/sec)
  - Control smooth muscle activity, heart rate, and glandular function

C Fibers

- **C Fibers:** Unmyelinated, Slow-Conducting, Found mainly in the autonomic nervous system
  - Small diameter and lack of myelination lead to slow conduction (0.5-2 m/sec)
  - Involved in transmitting slow, chronic pain and temperature sensations
  - **C fibers** are also part of the autonomic nervous system **(e.g., postganglionic autonomic fibers)**
  - Function in **long-lasting dull pain** and visceral sensations

Conduction Velocity and Myelination

- **The Role of Myelination in Nerve Conduction**
  - **Myelination** speeds up signal transmission
    - Increases membrane **resistance** and lowers **capacitance**
  - **Myelinated fibers** (A and B) use **saltatory conduction**, where the impulse jumps between nodes of Ranvier
  - **Unmyelinated fibers** (C) conduct more **slowly** due to continuous conduction

Conduction Velocity and Myelination, Cont.

- An axon's myelin sheath **is not** continuous.
- Every 1 to 2 mm is a 2- to 3-μm segment of exposed axonal membrane known as a **node of Ranvier**.
- **Nodes** are tightly packed with **Na^+ channels**, whereas the **internodal regions** (areas lying hidden beneath the myelin sheath) have virtually **no** channels.
- In practice, this means that an **action potential** leapfrogs from **one node to the next down the length of the axon**, a behavior known as **nodal**, or **saltatory conduction.**
- Breaks in the **myelin sheath** occur at the **nodes of Ranvier**.

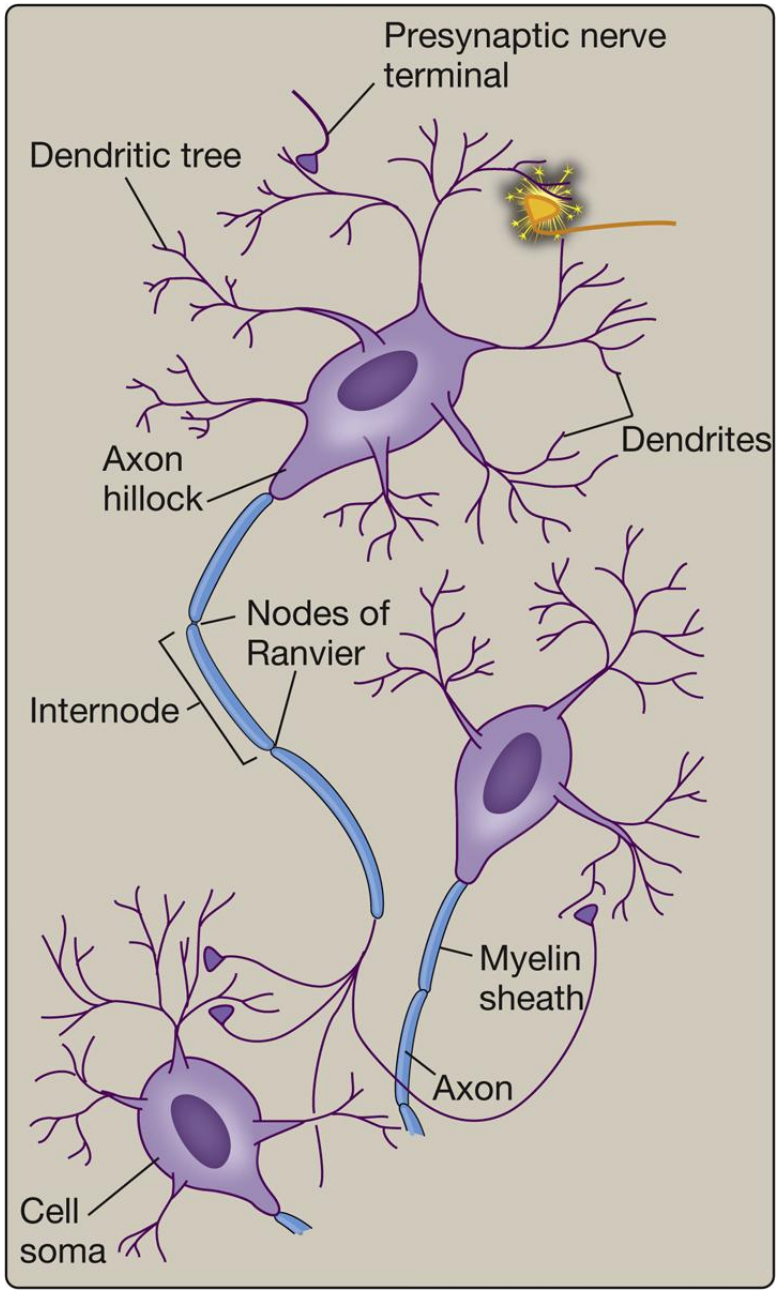


Figure 5.2 Neuronal anatomy. ***(Adopted from Lippincott Illustrated Reviews Physiology, 3rd Ed.)***

Conduction Velocity and Myelination, Cont.

- Breaks in the **myelin sheath** occur at the **nodes of Ranvier**.

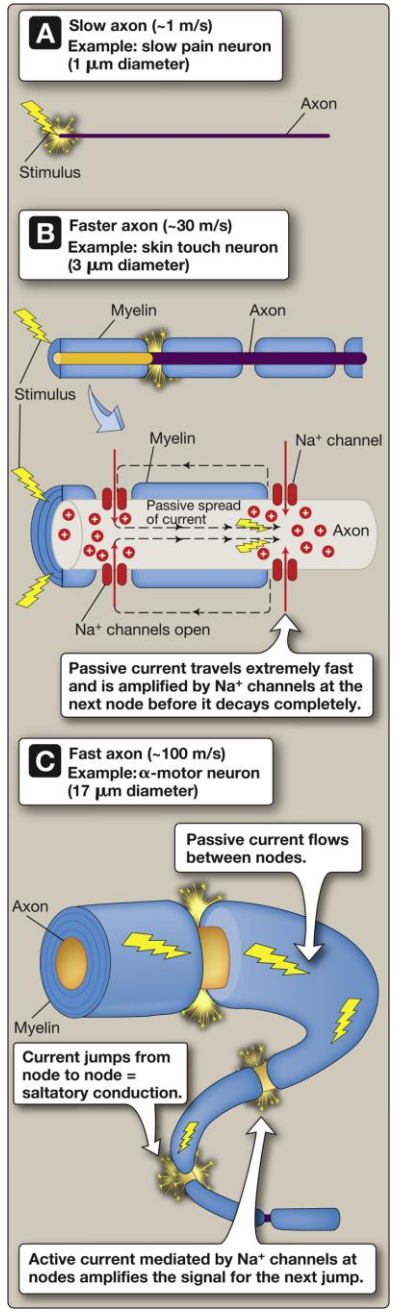


Figure 5.4 Myelin and diameter effects on axonal conduction velocity. ***(Adopted from Lippincott Illustrated Reviews Physiology, 3rd Ed.)***

Axonal Diameter and Conduction Velocity

- **The Role of Myelination in Nerve Conduction**
  - **Myelination** speeds up signal transmission
    - Increases membrane **resistance** and lowers **capacitance**
  - **Myelinated fibers** (A and B) use **saltatory conduction**, where the impulse jumps between nodes of Ranvier
  - **Unmyelinated fibers** (C) conduct more **slowly** due to continuous conduction

TABLE 3.1 Classification of Nerve Fibers

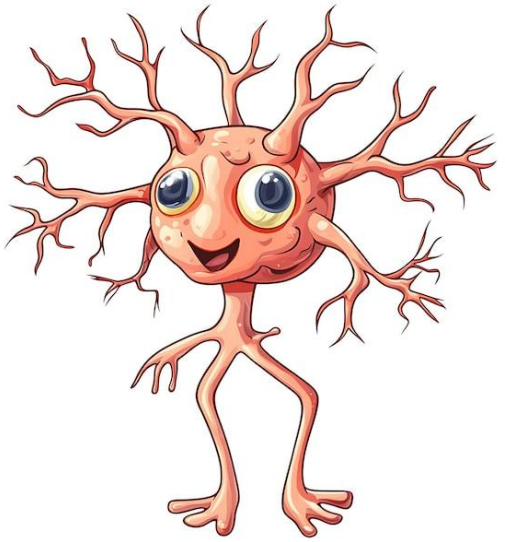
Classification	Type of Nerve Fiber	Example	Relative Diameter	Relative Conduction Velocity	Myelination
Sensory and Motor	Aα	α-Motoneurons	Largest	Fastest	Yes
	Aβ	Touch, pressure)	Medium	Medium	Yes
	AY	Y-motoneurons to muscle spindles (intrafusal fibers)	Medium	Medium	Yes
	AΔ	Touch, pressure, temperature, fast pain	Small	Medium	Yes
	B	Preganglionic autonomic nerves	Small	Medium	Yes
	C	Slow pain, postganglionic autonomic nerves; olfaction	Smallest	Slowest	No
Sensory only	Ia	Muscle spindle afferents	Largest	Fastest	Yes
	Ib	Golgi tendon organ afferents	Largest	Fastest	Yes
	II	Secondary afferents of muscle spindles, touch, pressure	Medium	Medium	Yes
	III	Touch, pressure, fast pain, temperature	Small	Medium	Yes
	IV	Pain, temperature, olfaction	Smallest	Slowest	No

Clinical Relevance

- **Disorders Related to Nerve Fibers**
  - **Peripheral Neuropathy:** Damage to peripheral nerves affecting **sensory and motor functions**.
  - **Multiple Sclerosis:** Damage to the myelin sheath of **A and B fibers** leading to **impaired transmission**.
  - **Diabetic Neuropathy:** **Affects C fibers**, causing **chronic pain** and **loss of sensation**.

Summary of Nerve Fiber Types

- **Nerve fibers** in the peripheral nervous system are classified according to their **diameter.**
- This is closely linked to the speed with which impulses spread along them (i.e., their **conduction velocity**, and allows fibers to be crudely characterized as to **function**.
- The **fastest fibers** are the **myelinated large-diameter neurons (Aα).**
- The **slowest fibers** are **unmyelinated small (C) fibers**
  - Transmit pain impulses



Sensory Reception

Types of Sensory Receptors

- **Sensory receptors** are specialized **cells** or **structures** that transduce environmental energy into electrical signals. **They are classified by modality (type of stimulus detected):**
  - **Mechanoreceptors** – respond to mechanical deformation (touch, pressure, vibration, stretch)
    - **Examples:** Pacinian corpuscles, Meissner's corpuscles, hair cells in cochlea
  - **Thermoreceptors** – detect temperature (warm/cold receptors in skin)
  - **Nociceptors** – pain receptors; detect tissue damage (mechanical, thermal, chemical)
  - **Photoreceptors** – detect light (rods and cones in retina)
  - **Chemoreceptors** – detect chemicals (taste buds, olfactory receptors, carotid body)
  - **Electromagnetic receptors** – specialized type (photoreceptors fall here)
  - **Proprioceptors** – detect body position and movement (muscle spindles, Golgi tendon organs)

Mechanoreceptors: Sensing Touch and Pressure

- Found in the skin and other organs.
- Activated by pressure or changes in pressure.
- The physical deformation of the receptor membrane opens ion channels.
- **Example: Pacinian corpuscles** in subcutaneous tissue respond to deep pressure; **Meissner corpuscles** in nonhairy skin (touch), **baroreceptors** in the carotid sinus (blood pressure), **hair cells** on the organ of Corti (audition) and in the semicircular canals (vestibular system).
- Once activated, the receptor generates an action potential that travels to the brain.

Thermoreceptors: Detecting Temperature

- Found in the skin.
- Activated by temperature or changes in temperature (detecting hot and cold stimuli).
- Specific receptors (e.g., TRPV1 for heat and TRPM8 for cold).
- Receptors activate ion channels, generating action potentials.
- Signals are transmitted to the brain where temperature changes are perceived.

Nociceptors: Pain Receptors

- Activated by extremes of pressure, temperature, or noxious chemicals.
- Activation leads to pain sensation to protect the body from injury.
- **Nociceptors** generate action potentials that travel to the spinal cord and brain.
- Chronic pain involves long-term activation of nociceptors.

Photoreceptors: Light Detection and Vision

- Located in the retina (rods and cones).
- Activated by light and are involved in vision.
- Rods detect low light, while cones detect color.
- **Phototransduction:** Light is absorbed, causing a change in retinal pigment.
- This change activates a signal cascade leading to action potentials sent to the visual cortex.

Chemoreceptors: Sensing Chemicals (Taste & Smell)

- Activated by chemical stimuli like taste (gustation) and smell (olfaction)
  - Gustatory cells in taste buds respond to specific chemicals (e.g., salty, sweet)
  - Olfactory receptors detect airborne molecules and send signals to the brain
- **Mechanism:** Chemical binds to receptors, initiating signal transduction pathways.

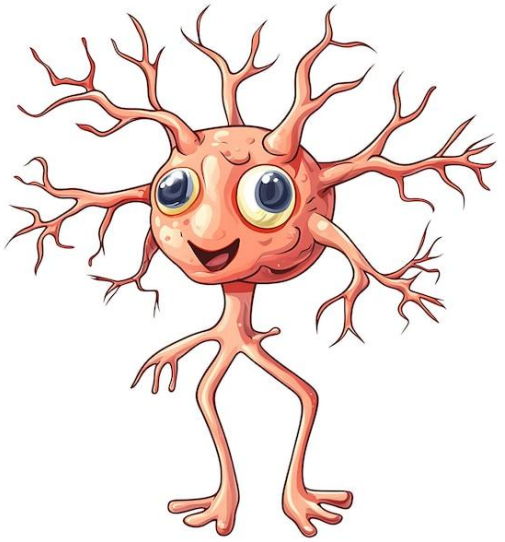
TABLE 3.2 Types and Examples of Sensory Receptors

Type of Receptor	Modality	Receptor	Receptor
Mechanoreceptors	Touch	Pacinian corpuscle	Skin
	Audition	Hair cell	Organ of Corti
	Vestibular	Hair cell	Macula, semicircular canal
Photoreceptors	Vision	Rods and cones	Retina
Chemoreceptors	Olfaction	Olfactory receptors	Olfactory mucosa
	Taste	Taste buds	Tongue
	Arterial PO_2		Carotid and aortic bodies
	pH and CSF		Ventrolateral medulla
Thermoreceptors	Temperature	Cold receptors	Skin
		Warm receptors	Skin
Nociceptors	Extremes of pain and temperature	Thermal nociceptors	Skin
		Polymodal nociceptors	Skin

CSF, Cerebrospinal fluid; ***PO_2***, partial pressure of oxygen.

Disorders Involving Visceral Sensory Neurons

- Visceral pain syndromes
- Autonomic dysreflexia
- Conditions like irritable bowel syndrome (IBS)
- **Impact on Homeostasis:** Malfunctioning of visceral sensory neurons can disrupt homeostasis and lead to chronic disorders.



Sensory Transduction and Receptor Potentials

Sensory Transduction

- **Sensory Transduction:** The process by which a **sensory receptor converts an external stimulus** (e.g., mechanical, thermal, chemical, or light energy) into an **electrical signal** that the nervous system can interpret.
- **Mechanism:**
  - Stimulus energy activates **ion channels** (directly or via second messengers)
  - This changes **ion fluxes (Na^+, K^+, Ca^{2+}, Cl^-)** across the receptor membrane

Receptor Potentials

- The **direct electrical result of sensory transduction**.
- A **graded, local change in membrane potential** in the receptor cell caused by the stimulus.
- **Properties:**
  - **Amplitude** depends on stimulus strength
  - Can be **depolarizing** (e.g., mechanoreceptors) or **hyperpolarizing** (e.g., rods in the retina)
  - **If threshold is reached**, the receptor potential initiates **action potentials** in the afferent neuron (or regulates neurotransmitter release in secondary synaptic pathways)

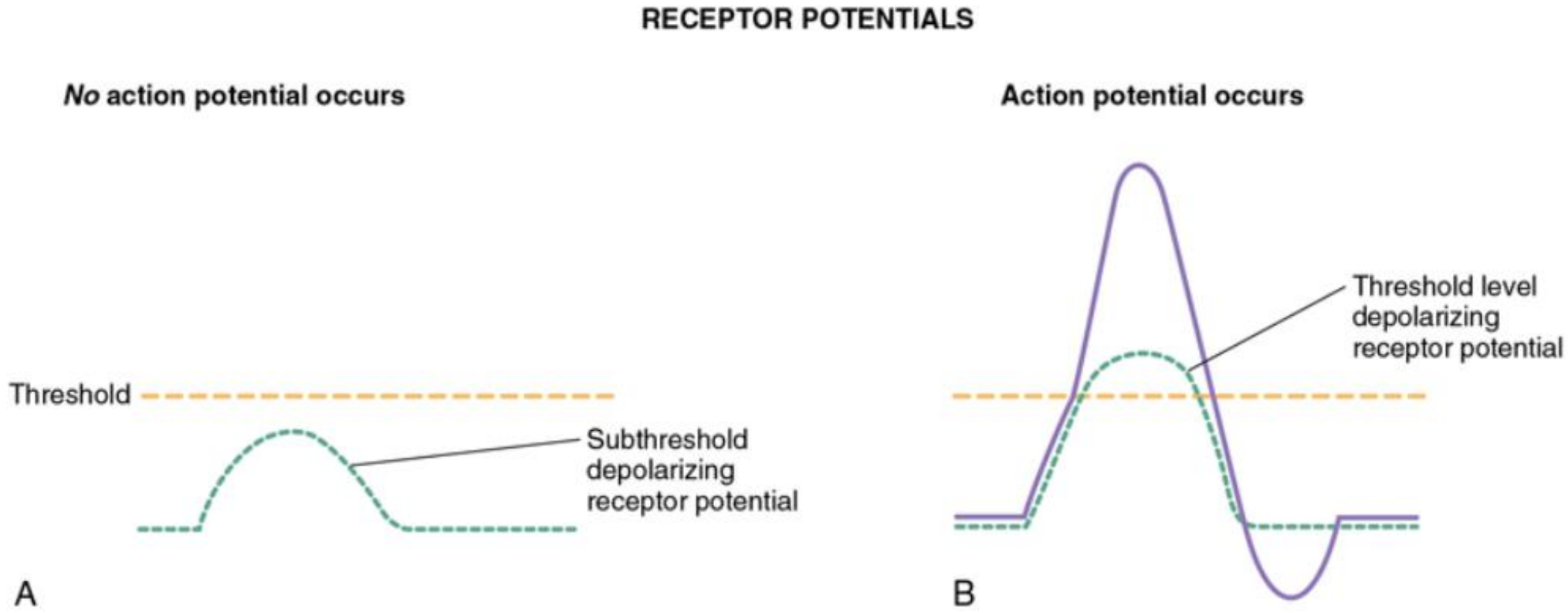


Figure. 3.5 Receptor potentials in sensory receptor cells. **Receptor potentials** may be either **depolarizing** (shown) or **hyperpolarizing** (not shown). **A, if a depolarizing receptor potential *does not*** bring the membrane potential to threshold, **no** action potential occurs. **B,** If a depolarizing receptor potential brings the membrane to threshold, then an **action potential** occurs in the sensory receptor.

Receptor Potentials, Cont.

- **Key properties:**
  - **Local and graded:** confined to receptor membrane, amplitude varies with stimulus
  - **Summation possible:** multiple stimuli can add together
    - **Decremental:** spreads passively and decreases with distance
  - **Threshold function:** must reach a certain level at the trigger zone to generate action potentials

Relationship Between the Two

- **Sensory transduction** is the **biophysical/biochemical process** (converts stimulus energy → electrical change).
- **Receptor potential** is the **product** (the graded potential generated by that process).
- **Put another way:**
  - **Transduction = cause** (conversion of stimulus energy)
  - **Receptor potential = effect** (graded electrical signal produced)

Example Pathways

- **Pacinian Corpuscle (touch)**
  - Mechanical deformation → mechanosensitive Na^+ channels open **(sensory transduction)** → depolarizing **receptor potential** → action potentials in afferent nerve
- **Rod Photoreceptor (vision)**
  - Photon absorption → rhodopsin activates transducin → PDE lowers cGMP → Na^+ channels close **(sensory transduction)** → hyperpolarizing **receptor potential** → reduced glutamate release to bipolar cells

Receptor Potentials in Pacinian Corpuscle

- **Pacinian Corpuscle (mechanoreceptor)**
  - **Structure:** Onion-like layers around a central nerve ending.
  - **Stimulus:** Vibration or pressure compresses the lamellae.

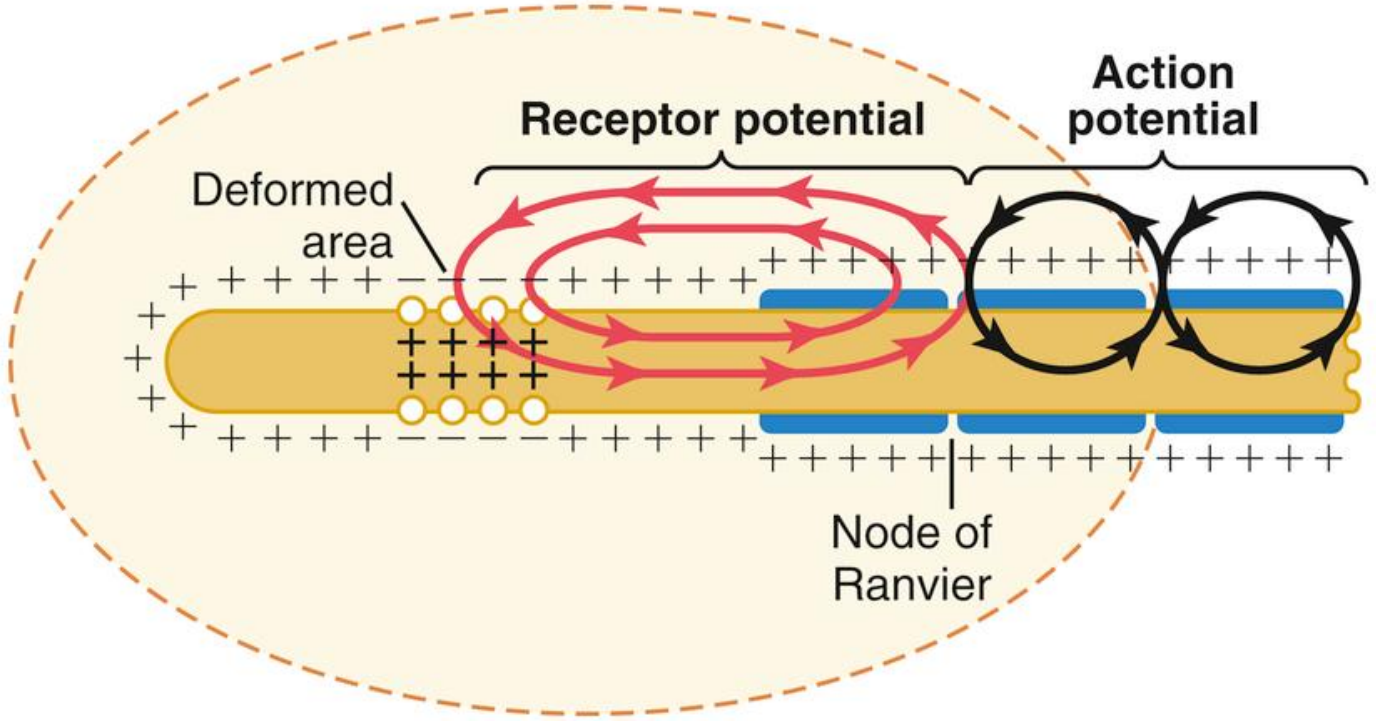


FIGURE 47-3. Excitation of a sensory nerve fiber by a receptor potential produced in a Pacinian corpuscle. (Adopted from *Guyton and Hall Textbook of Medical Physiology*, 13th ed.)

Receptor Potentials in Pacinian Corpuscle, Cont.

- **Mechanism:**
  - Mechanical deformation **(e.g., pressure)** stretches the nerve ending's membrane
  - This opens **mechanosensitive Na^+ channels**.
  - Na^+ influx (through mechanically gated ion channels)→ **local graded depolarization (receptor potential)**
  - If the receptor potential reaches threshold at the **first node of Ranvier**, voltage-gated Na^+ channels open
  - **Action potentials** are generated and propagated along the sensory fiber
- **Key point:** Pacinian corpuscles adapt rapidly (phasic receptor) → detect vibration and rapid changes, **not** steady pressure.

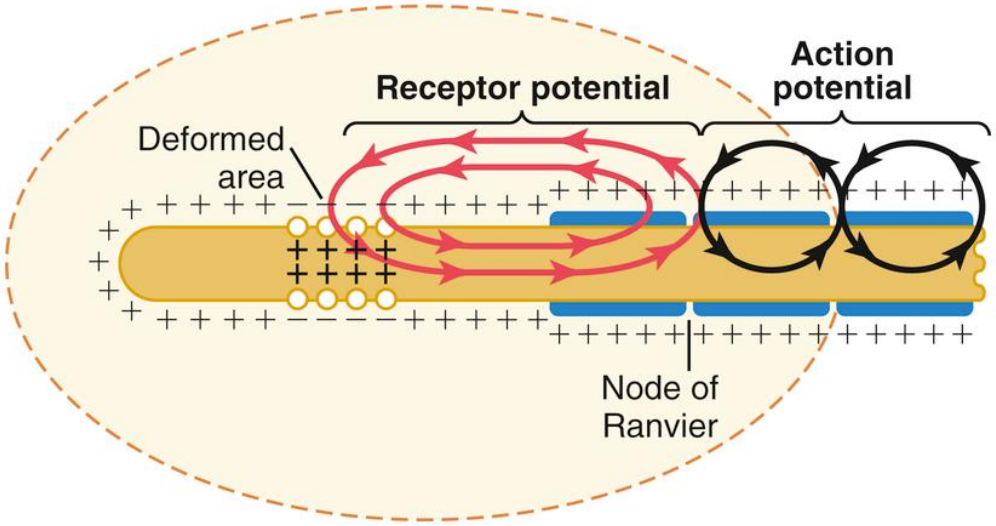


FIGURE 47-3. Excitation of a sensory nerve fiber by a receptor potential produced in a Pacinian corpuscle. (Adopted from *Guyton and Hall Textbook of Medical Physiology*, 13th ed.)

Receptor Potentials in Rod Photoreceptor

- **Rod Cell (photoreceptor)**
  - **Structure:** Outer segment contains rhodopsin (light-sensitive pigment).
  - **Stimulus:** Light (photons).

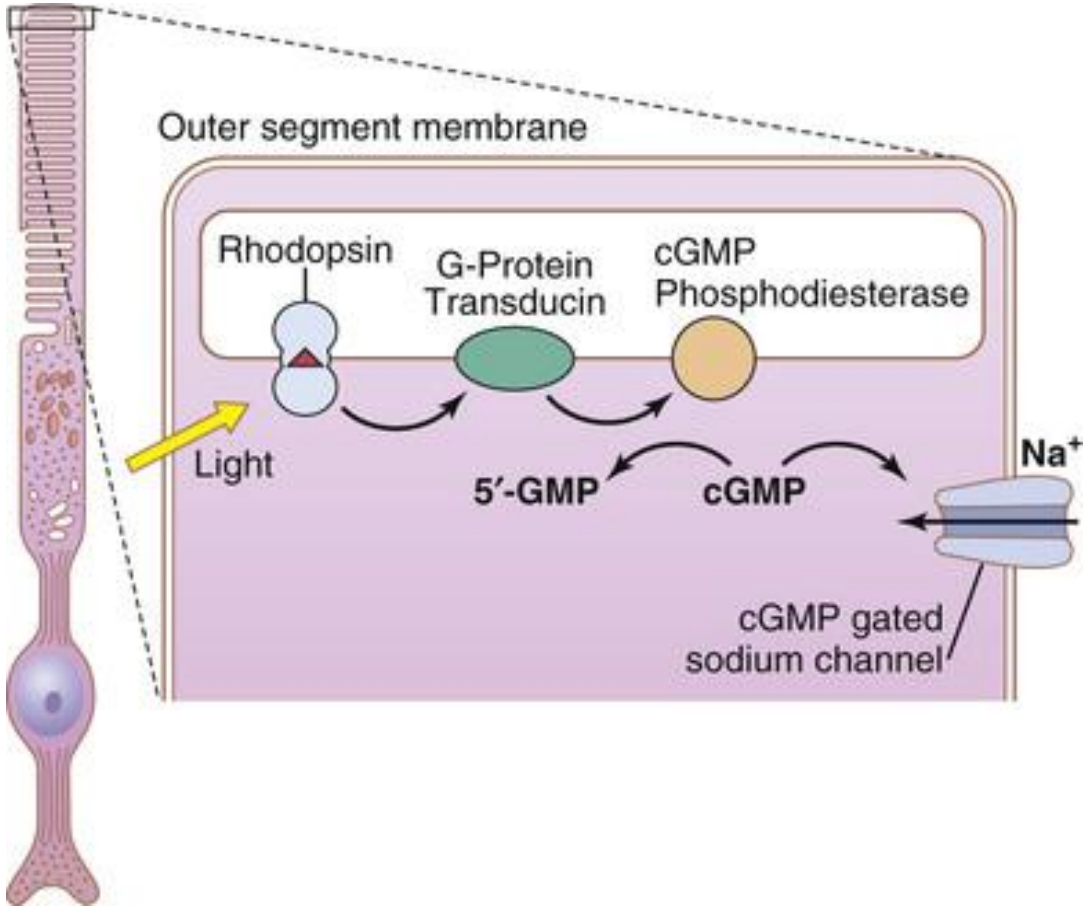


Figure 51-7 Phototransduction in the outer segment of the photoreceptor (rod or cone) membrane. Adopted from Guyton and Hall **Textbook of Medical Physiology**, 13th ed.

- **Mechanism:**
  - Light activates **rhodopsin → retinal changes shape → activates transducin (G-protein)**
  - Transducin activates **phosphodiesterase (PDE),** which decreases intracellular **cGMP**
  - Low cGMP → closure of **cGMP-gated Na^+/Ca^{2+} channels** in the outer segment membrane
  - Na^+ influx decreases → cell **hyperpolarizes** (unlike most receptors, rods hyperpolarize to stimulus)
  - Hyperpolarization reduces **glutamate release** at the synapse with bipolar cells
  - This modulates activity of bipolar → ganglion cells → optic nerve **action potentials**
- **Key point:** Rods and cones do not fire action potentials; they use graded potentials to regulate transmitter release.

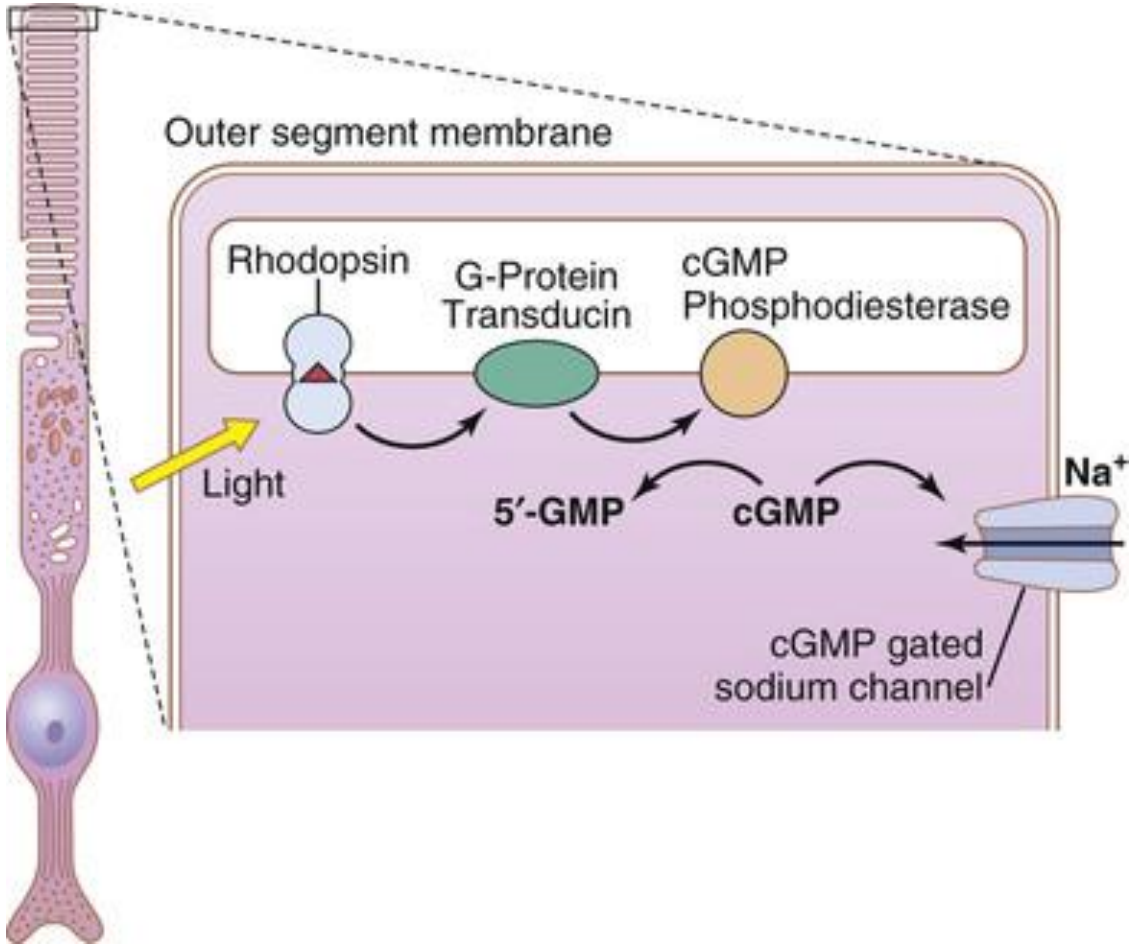


Figure 51-7 Phototransduction in the outer segment of the photoreceptor (rod or cone) membrane. Adopted from Guyton and Hall **Textbook of Medical Physiology**, 13th ed.

TABLE 1. Comparison: Pacinian Vs. Rod

Feature	Pacinian Corpuscle	Rod Cell (Photoreceptor)
Stimulus	Mechanical deformation (vibration)	Light (photons)
Ion channel type	Mechanosensitive Na^+ channels	cGMP-gated Na^+/Ca^{2+} channels
Response	Depolarization (generator potential)	Hyperpolarization (graded potential)
Signal encoding	Action potential in afferent nerve	Neurotransmitter release modulation
Adaptation	Rapid (phasic)	Sustained (tonic, but light-adaptive)

- **Sensory receptors** transduce environmental stimuli into electrical signals.
- **Pacinian corpuscle:** mechanical stimulus → depolarizing receptor potential → action potentials.
- **Rod cell:** light stimulus → hyperpolarizing receptor potential → decreased neurotransmitter release → downstream action potentials in ganglion cells.

Types of Sensory Receptors and Their Transduction Mechanisms

Mechanoreceptors

- **Modalities:** Touch, vibration, pressure, proprioception, hearing, balance
- **Stimulus:** Mechanical deformation (stretch, vibration, movement of hair cells).
- **Mechanism:**
  - Mechanical force opens **mechanosensitive ion channels** → Na^+ or K^+ influx
  - Generates a depolarizing receptor potential
- **Examples:**
  - **Pacinian corpuscles:** detect vibration/pressure via deformation of lamellae
  - **Hair cells (cochlea, vestibular system):** bending of stereocilia opens **tip-link gated K^+ channels** → depolarization

Thermoreceptors

- **Stimulus:** Temperature changes.
- **Mechanism:**
  - Heat or cold alters the gating of **transient receptor potential (TRP) ion channels**
  - Ion flux (Na^+/Ca^{2+}) → depolarizing receptor potential
- **Examples:**
  - **TRPV1:** activated by heat >43°C and capsaicin
  - **TRPM8:** activated by cold and menthol

Nociceptors

- **Modalities:** Pain
- **Stimulus:** Mechanical, thermal, or chemical injury.
- **Mechanism:**
  - Noxious stimuli activate ion channels (TRP family, acid-sensing ion channels, ATP-sensitive channels)
  - Depolarization generates action potentials in **Aδ** (fast pain) or **C fibers** (slow pain)

Photoreceptors

- **Modalities:** Vision
- **Stimulus:** Light (photons).
- **Mechanism:**
  - Photon absorption by **rhodopsin (rods) or photopsins (cones)** → activates **transducin (G-protein)**
  - Transducin activates **PDE**, reducing cGMP
  - **cGMP-gated Na^+/Ca^{2+} channels close** → membrane **hyperpolarization**
  - This decreases **glutamate release**, modulating bipolar/ganglion cell activity
- **Key point:** Photoreceptors hyperpolarize (opposite to most receptors).

Chemoreceptors

1. **Olfactory receptors** (smell)
   - Odorant molecules bind **GPCRs** in olfactory neurons
   - Activates **adenylyl cyclase** → **cAMP increase**
   - cAMP opens **cation channels** → depolarization
2. **Taste receptors** (gustation)
   - **Salt:** Na^+ influx directly through ENaC channels
   - **Sour:** H^+ entry blocks K^+ channels, depolarizing cell
   - **Sweet, bitter, umami:** GPCRs activate second messenger pathways → depolarization and neurotransmitter release

Chemoreceptors, Cont..

3. **Carotid body chemoreceptors**
   - Detect $\downarrow O_2$, $\uparrow CO_2$, $\downarrow$pH
   - Ion channel modulation leads to depolarization and neurotransmitter release onto afferent nerve endings

Proprioceptors

- **Modalities:** Muscle length, tension, position
- **Stimulus:** Stretch or tension in muscles/tendons.
- **Mechanism:**
  - Muscle spindle stretch → opens stretch-sensitive ion channels in **Ia afferents**
  - Golgi tendon organ tension → activates mechanosensitive channels in **Ib afferents**
  - Leads to **depolarizing receptor potentials** → action potentials in afferent fibers

Encoding Stimuli into Neural Signals

- **Amplitude of receptor potential =** proportional to stimulus intensity.
- **If threshold reached:** frequency of action potentials encodes intensity.
- **Adaptation:**
  - **Phasic receptors** (e.g., Pacinian corpuscle)**: rapid adaptation** → detect change.
  - **Tonic receptors** (e.g., muscle spindle)**: slow adaptation** → detect sustained input.

Summary

- Different sensory receptors transduce stimuli by **ion channel modulation:**
  - **Mechanoreceptors, thermoreceptors, nociceptors** → direct ion channel opening (Na^+, K^+, Ca^{2+})
  - **Photoreceptors** → GPCR cascade → closure of cGMP-gated channels → hyperpolarization
  - **Chemoreceptors** → GPCRs or direct ion channel modulation → depolarization.
  - **Proprioceptors** → mechanosensitive ion channels in muscle spindle or Golgi tendon organs
- All lead to a **receptor potential**, which, if large enough, drives **action potential** firing in sensory nerves or modulates neurotransmitter release.

Sensory Adaptation

Sensory Adaptation

- **Sensory adaptation** is the process by which sensory receptors **decrease their responsiveness** during a constant or unchanging stimulus
  - The receptor potential amplitude diminishes → fewer action potentials generated in the afferent nerve
  - This prevents the nervous system from being overloaded with redundant information

Types of Sensory Adaptation

- **Rapidly Adapting (Phasic) Receptors**
  - Respond strongly at the **onset and offset** of a stimulus
  - Stop firing during constant stimulation
  - Detect **changes, vibration, and movement**
  - **Examples: Pacinian corpuscles** (vibration), **Meissner's corpuscles** (touch)
- **Slowly Adapting (Tonic) Receptors**
  - Continue firing as long as the stimulus persists
  - Encode **duration and intensity** of the stimulus
  - **Examples: Merkel cells** (touch/pressure), **muscle spindles** (stretch), **nociceptors** (pain)

Physiological Mechanisms

- **Ion channel inactivation:** sustained stimulation causes receptor ion channels to gradually close (e.g., mechanosensitive Na^+ channels).
- **Receptor membrane properties:** viscoelastic deformation in mechanoreceptors (Pacinian corpuscle “slips” under constant pressure).
- **Synaptic adaptation:** in sensory pathways (e.g., decreased neurotransmitter release in olfactory neurons with continuous odor exposure).
- **Central adaptation:** modulation in higher CNS centers can also “tune down” repetitive signals.

Physiological Significance

1. **Efficiency of perception**
   - Frees the brain from constant background noise
   - Enhances sensitivity to **new or changing stimuli** (e.g., detecting sudden movement in vision or vibration in touch)
2. **Special senses examples**
   - **Vision:** retinal photoreceptors adapt to light levels (dark vs light adaptation)
   - **Smell (olfaction):** rapid adaptation prevents sensory overload in strong odors
   - **Touch:** you stop feeling your clothes after a while (Pacinian corpuscle adaptation)
3. **Homeostasis**
   - Prevents overstimulation by constant input, conserving energy for meaningful changes

Clinical Significance

1. **Loss of adaptation**
   - **Neuropathy** (e.g., diabetic neuropathy)**:** abnormal adaptation may impair detection of vibrations or changes in pressure, predisposing to foot ulcers
   - **Chronic pain syndromes:** nociceptors normally adapt slowly, but in neuropathic pain, adaptation fails → persistent, exaggerated pain sensation (hyperalgesia)
2. **Abnormal hypersensitivity**
   - **Tinnitus:** failure of auditory adaptation in cochlear or central circuits leads to persistent perception of sound
   - **Allodynia:** normally non-painful stimuli perceived as painful due to altered central adaptation

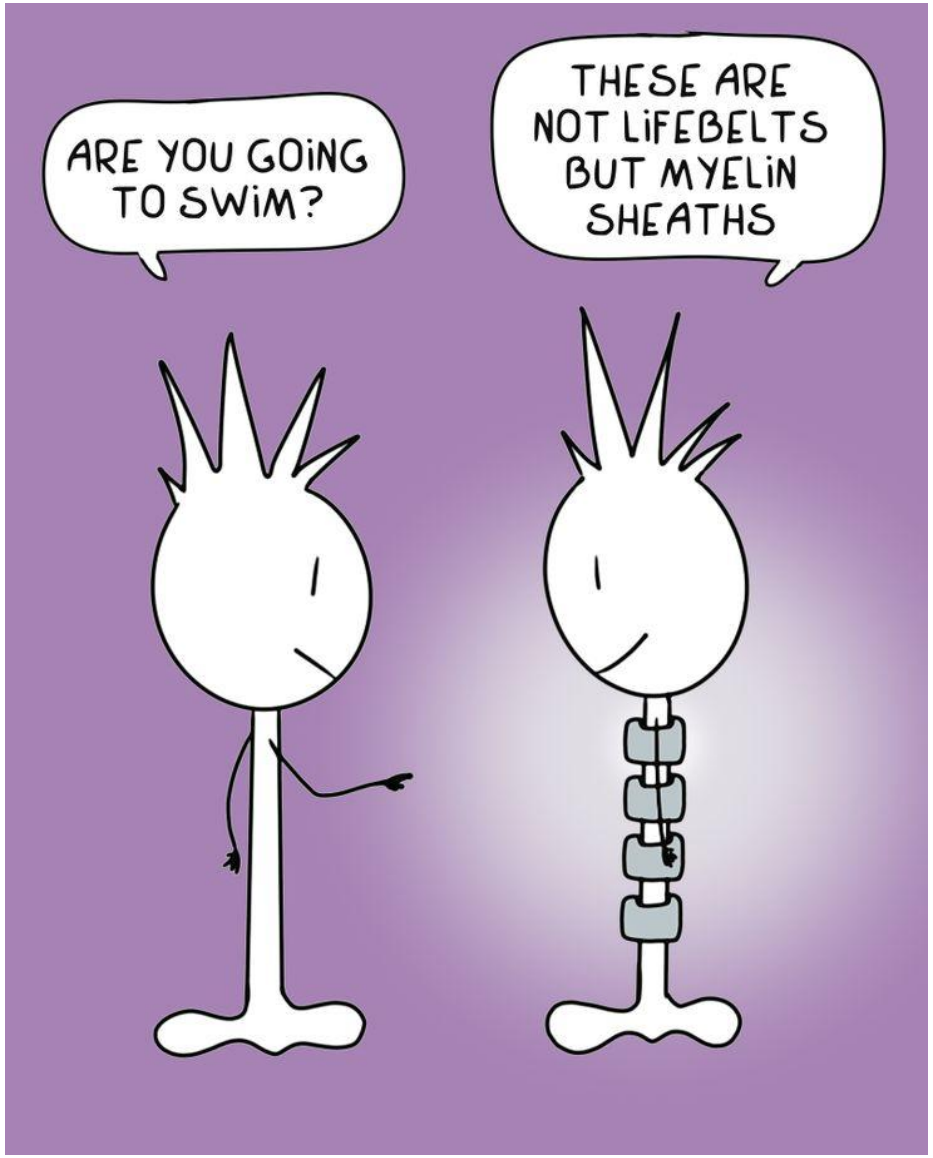
Clinical Significance, Cont.

3. **Deficits in sensory adaptation**
   - **Olfactory loss (anosmia/hyposmia):** impaired adaptation can affect appetite, safety (e.g., failure to detect smoke or gas)
   - **Autism spectrum disorders:** atypical sensory adaptation → hypersensitivity or hyposensitivity to sensory input

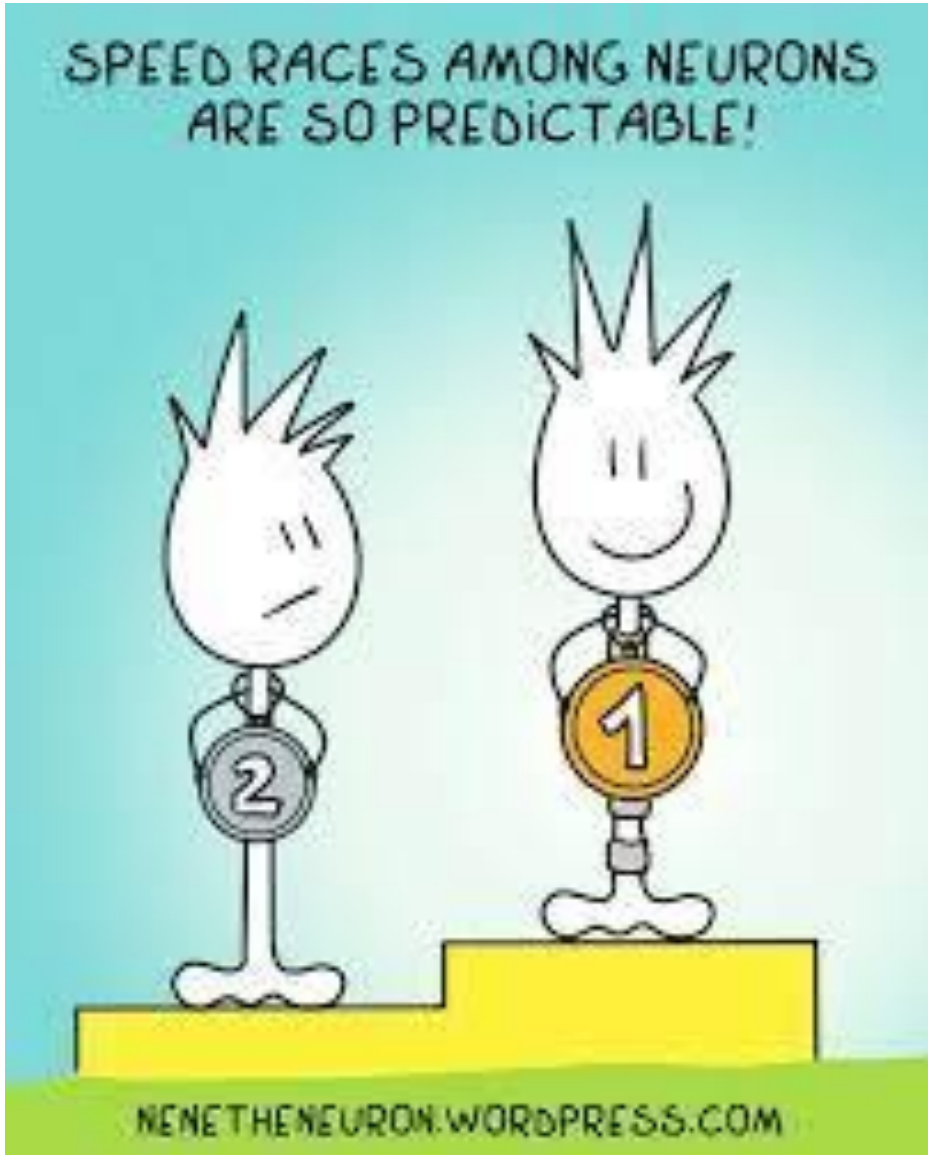
Summary

- **Sensory adaptation:** reduction in receptor response during sustained stimulation.
- **Rapid (phasic) receptors** → detect changes; **slow (tonic) receptors** → encode duration/intensity.
- **Physiological importance:** filters redundant signals, emphasizes new information.
- **Clinical relevance:** disrupted adaptation contributes to pain syndromes, sensory neuropathies, tinnitus, and sensory processing disorders.

The End !


ARE YOU GOING TO SWIM?
THESE ARE NOT LIFEBELTS BUT MYELIN SHEATHS

Questions !


SPEED RACES AMONG NEURONS ARE SO PREDICTABLE!
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Thank You !

